Evaluating the Efficiency and Accuracy of GPT-4o in Analyzing Pharmaceutical Annual Reports: A Comparative Study with Manual Review

Abstract

The rapid advancement of artificial intelligence (AI) technologies transforms data analysis, offering unprecedented speed and accuracy. This research project evaluates the efficiency and accuracy of GPT-4o, a cutting-edge AI language model, in analyzing annual reports of pharmaceutical companies. We manually review the 2023 annual reports of twenty pharmaceutical companies, taking approximately 8 hours and 30 minutes. In contrast, GPT-4 analyzes the same reports in an hour and a half, highlighting its potential to significantly enhance productivity.

We apply statistical tools to quantify efficiency gains and grade the accuracy of GPT-4's responses. Our analysis reveals that GPT-4 is 86% highly accurate with the perspective of an error baseline of 5%, this accuracy disregards directional discrepancies. It then demonstrates 90% high accuracy for values less than actual values(-5% baseline) and 96% high accuracy for values greater than actual values(+5% baseline). By categorizing the results based on accuracy levels, we identify which types of questions yield the most reliable answers.

This research aims to reveal the strengths and limitations of GPT-4 in financial analysis, a comprehensive understanding of the capabilities of GPT-4 in professional and financial data analysis, ultimately offering valuable insights into its practical applications and guiding future enhancements in AI-assisted data analysis in the pharmaceutical industry.

Keywords: AI Accuracy, AI Efficiency, Comparative Analysis, Data Analysis, Error Analysis, Financial Reports, GPT-4o, Human-AI Collaboration, Pharmaceutical Annual Reports, Time Savings.

Introduction

The integration of artificial intelligence (AI) into financial data analysis represents a significant shift in how industries manage and interpret vast amounts of information. While traditional methods rely on manual reviews, the advent of AI, particularly models like GPT-4o, promises enhanced efficiency and accuracy. This study looks to answer a pertinent question: How fast, accurate, and reliable is AI—more specifically, ChatGPT's most recent update (GPT-4o)—in reviewing financial reports?

AI applications have been widely studied across various sectors, but there is not much data for understanding how AI, particularly GPT-4o, performs in the analysis of financial reports within the pharmaceutical industry. This is not merely an academic gap but a practical one, given the complexities and regulatory requirements in pharmaceutical financial reporting, which directly impact investment decisions, research funding, and ultimately, patient care.

The relevance of this research lies in its focus on oncology drug research, an area where accurate financial analysis is crucial for directing resources towards life-saving treatments. Inaccuracies in

financial data can lead to misallocated funding, delayed research, and potential setbacks in drug development. By investigating GPT-4o's efficiency and accuracy, this study not only contributes to academic knowledge but also addresses a real-world need for reliable AI tools that can enhance the speed and precision of financial analysis in a high-stakes industry. The methodology involves a detailed comparative analysis between human/manual data collection and AI-assisted data extraction.

Methodology

Selection of Companies and their Reports

The selection of pharmaceutical companies for this study was a critical step to ensure the relevance of the analysis. The focus was on companies involved in oncology drug research, given the high stakes and significant investments associated with this field. The Drugs@FDA database was chosen for identifying relevant pharmaceutical companies. This database is a comprehensive resource maintained by the U.S. Food and Drug Administration (FDA), providing information about approved drugs and companies involved in drug development and research. The primary criterion was that the companies must be engaged in oncology drug research. You may wonder why oncology was chosen. It is one of the most dynamic and high-investment areas of pharmaceutical research, offering a rich dataset for analysis. Companies with recent significant activities, such as new drug approvals or ongoing major clinical trials in oncology, were prioritized to ensure the data reflects current industry trends.

After the search and identification process within the Drugs@FDA database to identify companies that met the oncology focus criterion, 20 pharmaceutical companies were selected for the study. These companies represent a mix of large multinational corporations and specialized biotech firms, providing a diverse sample of the industry. We cross-referenced additional sources such as the SEC EDGAR database(a federal website on company fillings available to the public eye), company websites, recent press releases, and industry reports to search for the most recent annual reports of the selected pharmaceutical companies, covering the fiscal year ending December 31, 2023 to ensure that the analysis was based on the latest available data, reducing the risk of outdated or irrelevant information since GPT-4o is up to date on data prior to October 2023. After the companies had been selected, I uploaded their individual documents in the model’s conversation focusing on extracting six critical financial metrics from each report.

Financial Prompt Topics

a. Total Revenue/Net Sales: The total income generated from the company's core business operations.
b. Operating Profits/Loss: The profit or loss from the company's primary business activities, excluding non-operating income and expenses.
c. Earnings Per Share (EPS): The portion of a company's profit allocated to each outstanding share of common stock.

d. Research and Development (R&D): Expenditures related to the company's investment in new product development and innovation.
e. Total Employees: The number of employees working for the company, indicating its operational scale.
f. Operating Margin Percentage: A measure of profitability calculated as operating income divided by net sales, expressed as a percentage. If not explicitly stated in the reports, this was calculated manually using the provided data.

Data Analysis Procedure

The data analysis procedure in this study involved a detailed comparison of the data collected manually and the data generated by GPT-4o. The following metrics and calculations were used to achieve this goal:

g. Speed Ratio which was calculated by comparing the time taken to manually collect data with the time taken by GPT-4o to generate the same data to highlight the efficiency gain achieved with GPT-4o.
h. Total errors were calculated by comparing each data point collected manually with the corresponding data point generated by GPT-4o. These errors were identified as the differences between the manually collected values and the AI collected values. It is important to note that this methodology assumes that the manual review is free of mistakes. However, this assumption should be explicitly stated and justified, acknowledging that while manual reviews are typically thorough and accurate, they are not entirely infallible. Therefore, the comparison primarily serves to highlight the AI's performance relative to a high standard of human accuracy.
i. Average, Median, and Standard Deviation of Errors to provide insight into the variability and consistency of GPT-4o's performance. These statistical measures were calculated for both the percentage errors and the absolute percentage errors to provide a detailed understanding of the AI's accuracy.
j. k-Percentiles of Error Calculation were calculated for the directional percentage errors to analyze the distribution of errors across all data points. Percentiles from 0th to 100th were determined, indicating the range and frequency of errors within the data set to identify how often GPT-4o's generated data deviated from the manually collected data and to what extent.
k. k-Percentiles of Absolute Error Calculation were calculated for the absolute percentage errors, providing a measure of accuracy without considering the direction of the error. Similar to the directional errors, percentiles from 0th to 100th were determined but focused solely on the magnitude of the errors to provide a clear understanding of the accuracy of GPT-4o's data generation, irrespective of whether the errors were positive or negative.

Results

Human Manual Review Process (Table 1)

Company Name	Total Revenue/Net Sales(M)	Operating Profits(M)	Basic Earnings Per Share	Research and Development (R&D) (M)	Total Employees	Operating Margin (%)
ASTAZENECA	45811	8193	3.84	10935	89900	17.90%
BRISTOL MYERS SQUIBB	45006	8440	3.86	9299	30000	18.75%
HOFFMANN-LA ROCHE (CHF)	58716	19240	9.6	13237	103605	32.77%
PFIZER	58496	1058	0.38	10679	88000	1.80%
TAKEDA PHARMS (yen)	4263800	214100	92	774391.2	50000	5.00%
EMD SERONO INC (euro)	20993	3609	6.49	2445	62908	17.20%
NOVARTIS PHARM	45440	9769	7.15	11371	76057	21.50%
BLUEPRINT MEDICINES	249.38	-507	1	427.72	655	-203.30%
ELI LILLY AND COMPANY	34124.1	6554.6	5.82	9313.4	43000	19.20%
MODERNA INC	6848	-4239	-12.33	4845	5600	-61.90%
GSK (pounds)	30328	6745	1.216	6223	70212	22.20%
AMGEN	28190	7897	12.56	4784	26700	28.01%
MILLENNIUM	265.76	2.687	0.0129	1	4887	1.01%
EXELIXIS INC	1830.208	170.885	0.65	1044.071	1310	9.34%
ONYX HEALTHCARE	1492.86	211.254	7.65	79.362	153	14.15%
REGENERON PHARMAS	13117.2	4047.1	37.05	4,439	13450	30.85%
DAIICHI SANKYO (yen)	1278478	120580	56.96	341570	17435	9.43%
BAYER (euro)	47637	612	-2.99	5371	99723	1.28%
BIOMARIN	2419.226	-158.097	-0.89	746.8	3401	-6.54%
EISAI (yen)	741800	53400	147.9	16900	1	7.20%

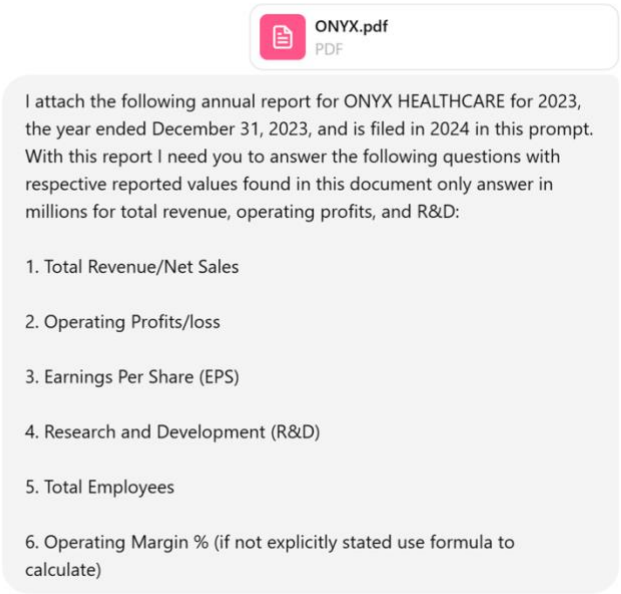
*The green highlight is for observations that didn't yield results (for accuracy calculations it is represented as 1)
*M = Millions
*CHF = Swiss Francs

The manual human review process took approximately 8 hours and 30 minutes to collect data on 6 key financial topics for each company, resulting in a total of 120 data points.

GPT-4o Prompt Entry

To minimize bias, we provided only one conversational input/prompt per company and started a new chat for each company discussion, The conversation prompt is as below:

AI Conversation Exchange Example:



Based on the ONYX HEALTHCARE annual report for the year ended December 31, 2023, here are the required values:

1. **Total Revenue/Net Sales**: NT$ 1,492.86 million【7:2†ONYX.pdf】.
2. **Operating Profits/Loss**: The report does not directly state operating profits or losses. However, net income for the period is NT$ 255.17 million【7:5†ONYX.pdf】.
3. **Earnings Per Share (EPS)**: NT$ 7.65 (Basic) and NT$ 7.59 (Diluted)【7:1†ONYX.pdf】.
4. **Research and Development (R&D)**: NT$ 79.36 million【7:1†ONYX.pdf】.
5. **Total Employees**: 153【7:0†ONYX.pdf】.
6. **Operating Margin %**:

   The operating margin is calculated as follows:

$$\text{Operating Margin} = \left(\frac{\text{Operating Income}}{\text{Total Revenue}}\right) \times 100$$

   Using the given net income and total revenue:

$$\text{Operating Margin} = \left(\frac{255.17}{1,492.86}\right) \times 100 \approx 17.1\%$$

AI (Table 2)

Company Name	Total Revenue/Net Sales(M)	Operating Profits(M)	Basic Earnings Per Share (EPS)	Research and Development	Total Employees	Operating Margin (%)
ASTAZENECA	45811	8193	3.84	10935	89900	17.90%
BRISTOL MYERS SQUIBB	45000	0	3.86	9300	30000	0.00%
HOFFMANN-LA ROCHE (CHF)	58716	19240	9.6	13237	103605	32.76%
PFIZER	58,496	1058	0.38	11407	83000	1.80%
TAKEDA PHARMS (yen)	4263800	214100	92	0	0	5.00%
EMD SERONO INC (euro)	20993	3609	6.49	2400	62908	17.20%
NOVARTIS PHARM	45440	9769	7.15	11371	76057	21.50%
BLUEPRINT MEDICINES	249.4	-506	1	427.7	655	-202.90%
ELI LILLY AND COMPANY	34124.1	6554.6	5.82	9313.4	43000	19.20%
MODERNA INC	6848	-4239	-12.33	4845	5600	-61.89%
GSK (pounds)	30300	6745	1.216	5562	70212	22.20%
AMGEN	28190	7897	12.56	4784	0	29.30%
MILLENNIUM	265.76	0.95	0.0129	1	4887	0.36%
EXELIXIS INC	1830.2	170.9	0.65	1044.1	1310	9.34%
ONYX HEALTHCARE	1492.86	255.17	7.65	79.36	153	17.10%
REGENERON PHARMAS	13117.2	4047.1	37.05	4439	13450	30.85%
DAIICHI SANKYO (yen)	1278500	120600	56.96	341600	17435	9.40%
BAYER (euro)	49253	612	-2.99	5371	99723	1.24%
BIOMARIN	2419.2	230.1	0.89	832.5	3401	9.50%
EISAI (yen)	741800	53400	152.5	16750	1	7.20%

*The yellow highlight is for observations that didn't yield results (for accuracy calculations it is represented as 0)
*The green highlight is for observations that didn't yield results and matched with the manual table results (for accuracy calculations it is representation as 1)
*M = Millions
*CHF = Swiss Francs

GPT-4o analyzed all 20 annual reports and extracted the required data in approximately 1 hour and 30 minutes. The extracted data was organized into an Excel spreadsheet, creating a table with six columns for the prompt questions and twenty rows for the companies.

Speed Analysis

The manual analysis process to review and record observations for the 20 companies took approximately 510 minutes (about 8 and a half hours) while the AI process took about an hour and 27 minutes. This means that for this research GPT-4ois about 5.86 times faster than manual data collection.

Next, for each of the 120 data observations, a percentage error in AI deviation was calculated by dividing the manually collected value by the corresponding AI-generated value. This comparison was organized into a separate table (Table 3).

Directional Inaccuracy Rate (Table 3)

The table below was created to view the measure of accuracy considering the direction of error deviations by comparing both my manual/human and AI observations(Table 3).

Company Name	Total Revenue/Net Sales	Operating Profits	Basic Earnings Per Share	Research and Development	Total Employees	Operating Margin
ASTAZENECA	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
BRISTOL MYERS SQUIBB	-0.01%	-100.00%	0.00%	0.01%	0.00%	-100.00%
HOFFMANN-LA ROCHE	0.00%	0.00%	0.00%	0.00%	0.00%	-0.03%
PFIZER	0.00%	0.00%	0.00%	6.82%	-5.68%	0.00%
TAKEDA PHARMS	0.00%	0.00%	0.00%	-100.00%	-100.00%	0.00%
EMD SERONO INC	0.00%	0.00%	0.00%	-1.84%	0.00%	0.00%
NOVARTIS PHARM	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
BLUEPRINT MEDICINES	0.01%	-0.20%	0.00%	0.00%	0.00%	-0.20%
ELI LILLY AND COMPANY	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
MODERNA INC	0.00%	0.00%	0.00%	0.00%	0.00%	-0.02%
GSK	-0.09%	0.00%	0.00%	-10.62%	0.00%	0.00%
AMGEN	0.00%	0.00%	0.00%	0.00%	-100.00%	4.61%
MILLENNIUM	0.00%	-64.64%	0.00%	0.00%	0.00%	-64.36%
EXELIXIS INC	0.00%	0.01%	0.00%	0.00%	0.00%	0.00%
ONYX HEALTHCARE	0.00%	20.79%	0.00%	0.00%	0.00%	20.85%
REGENERON PHARMAS	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
DAIICHI SANKYO	0.00%	0.02%	0.00%	0.01%	0.00%	-0.32%
BAYER	3.39%	0.00%	0.00%	0.00%	0.00%	-3.13%
BIOMARIN	0.00%	-245.54%	-200.00%	11.48%	0.00%	-245.26%
EISAI	0.00%	0.00%	3.11%	-0.89%	0.00%	0.00%

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Absolute Inaccuracy Rate (Table 4)

Another table was created to convert the percentage errors into absolute values using Excel's absolute formula, which provides a measure of accuracy without considering the direction of the error (Table 4).

Company Name	Total Revenue/Net Sales	Operating Profits	Basic Earnings Per Share	Research and Development	Total Employees	Operating Margin
ASTAZENECA	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
BRISTOL MYERS SQUIBB	-0.01%	-100.00%	0.00%	0.01%	0.00%	-100.00%
HOFFMANN-LA ROCHE	0.00%	0.00%	0.00%	0.00%	0.00%	-0.03%
PFIZER	0.00%	0.00%	0.00%	6.82%	-5.68%	0.00%
TAKEDA PHARMS	0.00%	0.00%	0.00%	-100.00%	-100.00%	0.00%
EMD SERONO INC	0.00%	0.00%	0.00%	-1.84%	0.00%	0.00%
NOVARTIS PHARM	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
BLUEPRINT MEDICINES	0.01%	-0.20%	0.00%	0.00%	0.00%	-0.20%
ELI LILLY AND COMPANY	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
MODERNA INC	0.00%	0.00%	0.00%	0.00%	0.00%	-0.02%
GSK	-0.09%	0.00%	0.00%	-10.62%	0.00%	0.00%
AMGEN	0.00%	0.00%	0.00%	0.00%	-100.00%	4.61%
MILLENNIUM	0.00%	-64.64%	0.00%	0.00%	0.00%	-64.36%
EXELIXIS INC	0.00%	0.01%	0.00%	0.00%	0.00%	0.00%
ONYX HEALTHCARE	0.00%	20.79%	0.00%	0.00%	0.00%	20.85%
REGENERON PHARMAS	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
DAIICHI SANKYO	0.00%	0.02%	0.00%	0.01%	0.00%	-0.32%
BAYER	3.39%	0.00%	0.00%	0.00%	0.00%	-3.13%
BIOMARIN	0.00%	-245.54%	-200.00%	11.48%	0.00%	-245.26%
EISAI	0.00%	0.00%	3.11%	-0.89%	0.00%	0.00%

Total Error Occurrences (Table 5)

Company Name	Total Revenue/Net Sales	Operating Profits	Basic Earnings Per Share	Research and Development	Total Employees	Operating Margin
ASTAZENECA	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%
BRISTOL MYERS SQUIBB	0.01%	100.00%	0.00%	0.01%	0.00%	100.00%
HOFFMANN-LA ROCHE	0.00%	0.00%	0.00%	0.00%	0.00%	0.03%
PFIZER	0.00%	0.00%	0.00%	6.82%	5.68%	0.00%
TAKEDA PHARMS	0.00%	0.00%	0.00%	100.00%	100.00%	0.00%
EMD SERONO INC	0.00%	0.00%	0.00%	1.84%	0.00%	0.00%
NOVARTIS PHARM	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
BLUEPRINT MEDICINES	0.01%	0.20%	0.00%	0.00%	0.00%	0.20%
ELI LILLY AND COMPANY	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
MODERNA INC	0.00%	0.00%	0.00%	0.00%	0.00%	0.02%
GSK	0.09%	0.00%	0.00%	10.62%	0.00%	0.00%
AMGEN	0.00%	0.00%	0.00%	0.00%	100.00%	4.61%
MILLENNIUM	0.00%	64.64%	0.00%	0.00%	0.00%	64.36%
EXELIXIS INC	0.00%	0.01%	0.00%	0.00%	0.00%	0.00%
ONYX HEALTHCARE	0.00%	20.79%	0.00%	0.00%	0.00%	20.85%
REGENERON PHARMAS	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
DAIICHI SANKYO	0.00%	0.02%	0.00%	0.01%	0.00%	0.32%
BAYER	3.39%	0.00%	0.00%	0.00%	0.00%	3.13%
BIOMARIN	0.00%	245.54%	200.00%	11.48%	0.00%	245.26%
EISAI	0.00%	0.00%	3.11%	0.89%	0.00%	0.00%

Key (%): (0<x<1) (1≤x<10) (10≤x<100) (x≥100)

In Table 5, this data analysis specifically looks at total error occurrences in absolution to provide a straightforward measure of accuracy without the need to consider the direction of the error. 47% of the errors (19 out of 40) which accounts for less than 1% of the inaccuracy of the AI observations. This indicates a high level of accuracy, as nearly half of the errors were minor deviations from the manually collected data. 18% of the errors (7 out of 40) accounts for $1\% \leq x < 10\%$ inaccuracy rate in the observations and shows moderate accuracy but still within acceptable limits for some applications. 15% of the errors (6 out of 40) which accounts for $10\% \leq x < 100\%$ inaccuracy which is very significant, these errors may indicate areas where GPT-4o needs improvement in handling specific types of data or reports. The remaining 20% of the errors (8 out of 40) are for inaccuracy rates that were greater than 100% error and incredibly substantial, indicating potential outliers or significant discrepancies that need further investigation. Due to most errors in the absolute percentage errors being minor, this analysis suggests that GPT-4o is generally reliable for extracting financial data from annual reports. However, the presence of significant errors also points to the need for more careful validation.

Directional Errors

The full sample analysis for the average error had a -10.60% which indicates that, on average, GPT-4o underestimated the values compared to the manually collected data. This negative average suggests a consistent tendency to report lower values across the sample. The directional median error across the sample resulted in 0.00% means that the middle value of the error distribution is zero, indicating that for more than half of the observations, GPT-4o's results were exactly matched to human/manual observations. A high standard deviation for the total sample was 41.66% which reveals significant variability in the errors, suggesting that while most predictions were accurate, others deviated substantially from the manual values.

Average Directional Error

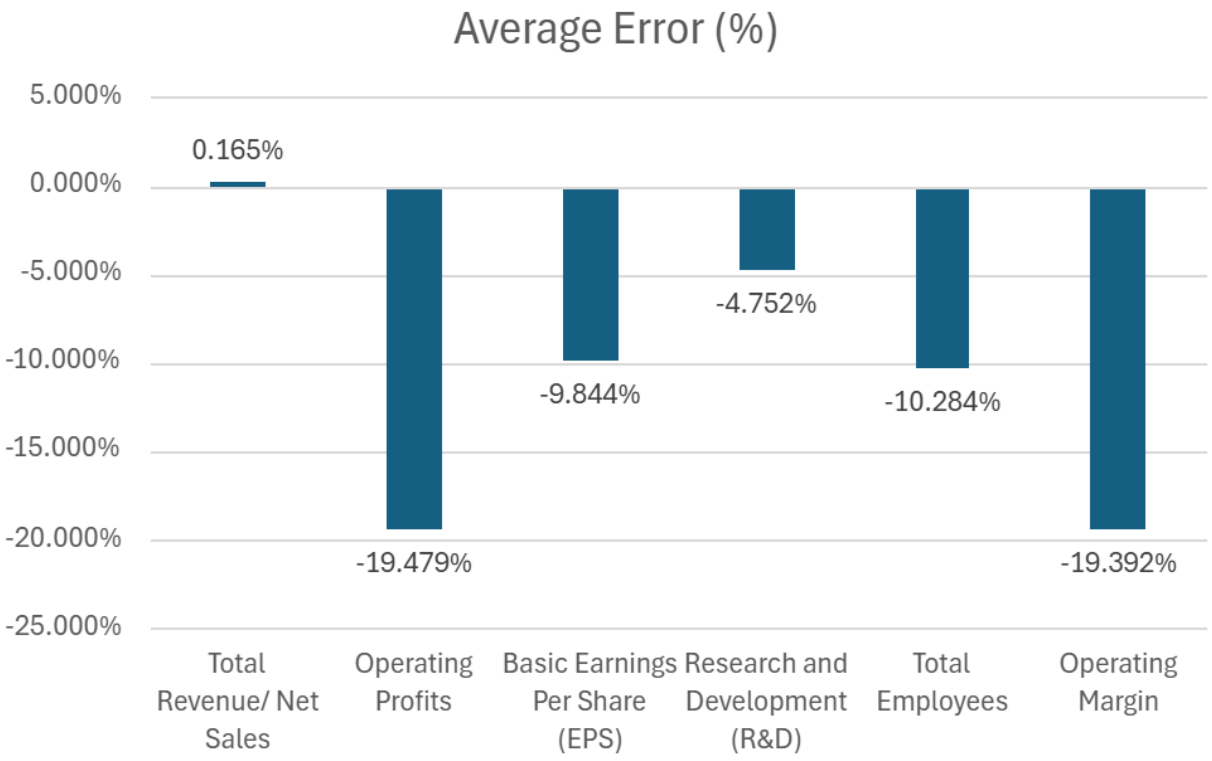


Fig. 2.

The figure above (fig. 2.) represents the average percentage error per prompt question comprising of the observations for the 20 companies used for our analysis. On the x-axis; we have the prompt question, and for our y-axis; there is the percentage from zero and upwards. The average for total revenue/net sales was 0.165% indicating GPT-4o's high accuracy in predicting total revenue/net sales, but on average greater than its actual values. In operating profits there is a -19.479%, this significant average error indicates that GPT-4o frequently underestimated operating profits, this direction is due to the negative value for the average error. The same is seen for operating margin at -19.392% followed by total employees (-10.284%), EPS (-9.844%) and lastly R&D (-4.752%).

Standard Deviation of Directional Error

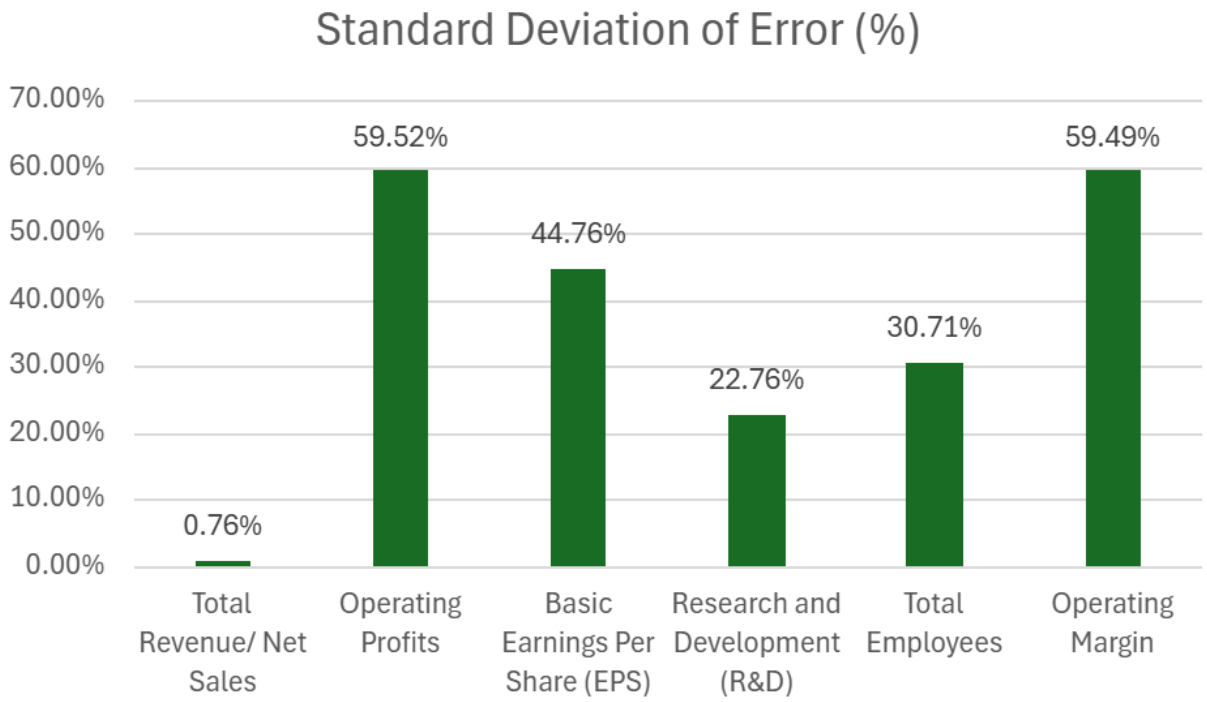


Fig. 3.

Similarities are shown in standard deviations of errors for respective prompt questions; Total Revenue/Net Sales (0.76%), Operating Profits (59.52%), EPS (44.76%), R&D (22.76%), Total Employees (30.71%), Operating Margin (59.49%). Although the average error rate of the basic earnings per share has -9.44%, it has a higher standard deviation of error (44.76%) than that of total employees which has a higher average error rate (-10.284%) which tell us that in the directional average error rate calculations, although results are similar in terms of average error rates of prompts linearly matching up with their standard deviations, is not directionally proportional. This could be due to significantly inaccurate values in specific prompt observations.

For the median of each prompt question results were as follows, Total Revenue/Net Sales (0.000%), Operating Profits (0.000%), EPS (0.00%), R&D (0.003%), Total Employees (0.000%), Operating Margin (0.008%). Zero median and near-zero median error confirms many exact matches.

Absolute Errors

The full sample analysis revealed an absolute average error of 11.78%, indicating that GPT-4o's predictions deviated from actual values by approximately 11.78% on average. The absolute median error was 0.00%, showing that for at least half of the observations, GPT-4o's predictions were very close to the manual data. However, the high standard deviation of 41.34% indicates significant variability in the errors, suggesting substantial deviations in some predictions.

Absolute Average Error

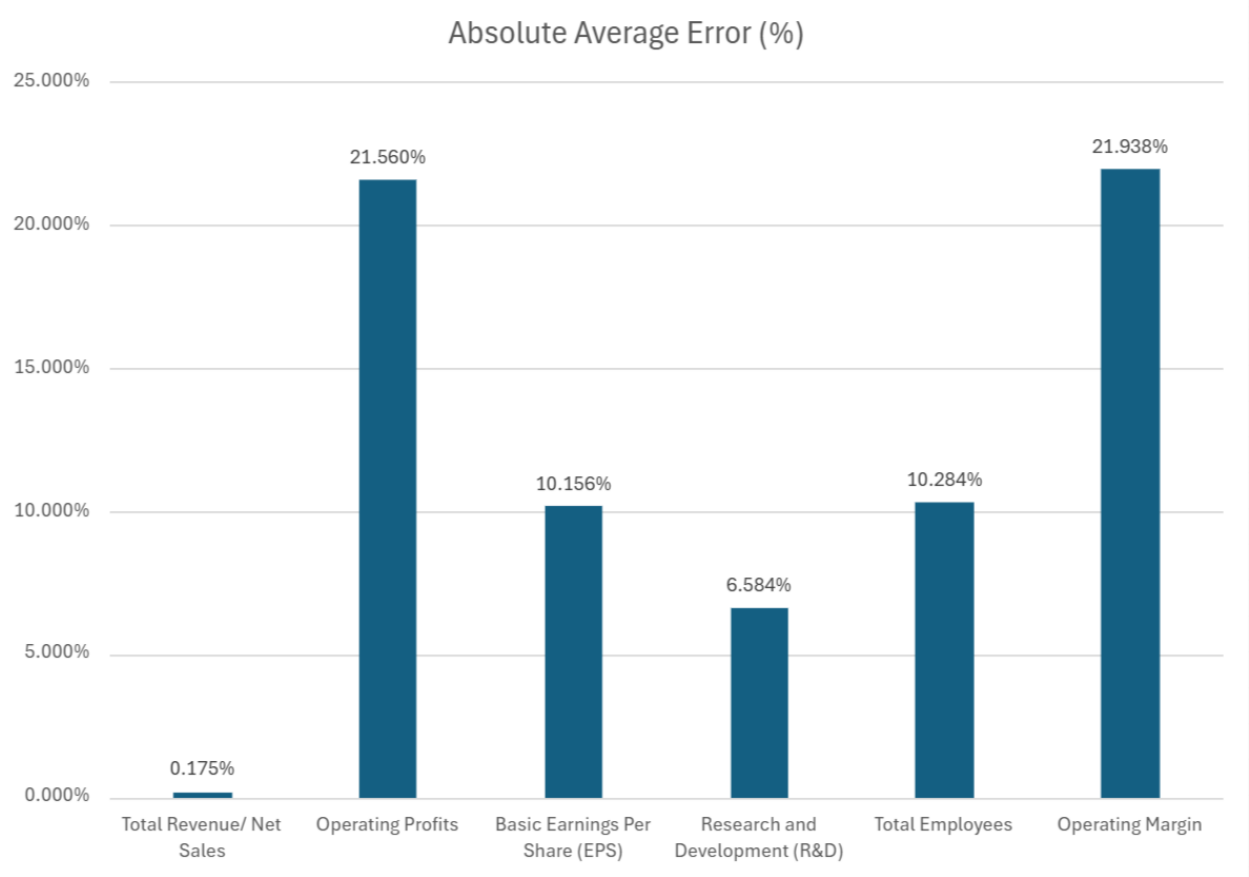


Fig. 4.

Figure 4 represents the absolute average percentage error per prompt question for the 20 companies analyzed. The x-axis shows the prompt questions, and the y-axis displays the percentage. The average error for total revenue/net sales was 0.175%, indicating GPT-4o's high accuracy in this area, though slightly above actual values. For operating margin, the absolute average error was 21.938%, showing significant deviation from actual values. Similar deviations were seen for operating profits (21.560%), total employees (10.284%), EPS (10.156%), and R&D (6.584%).

Standard Deviation of Absolute Error

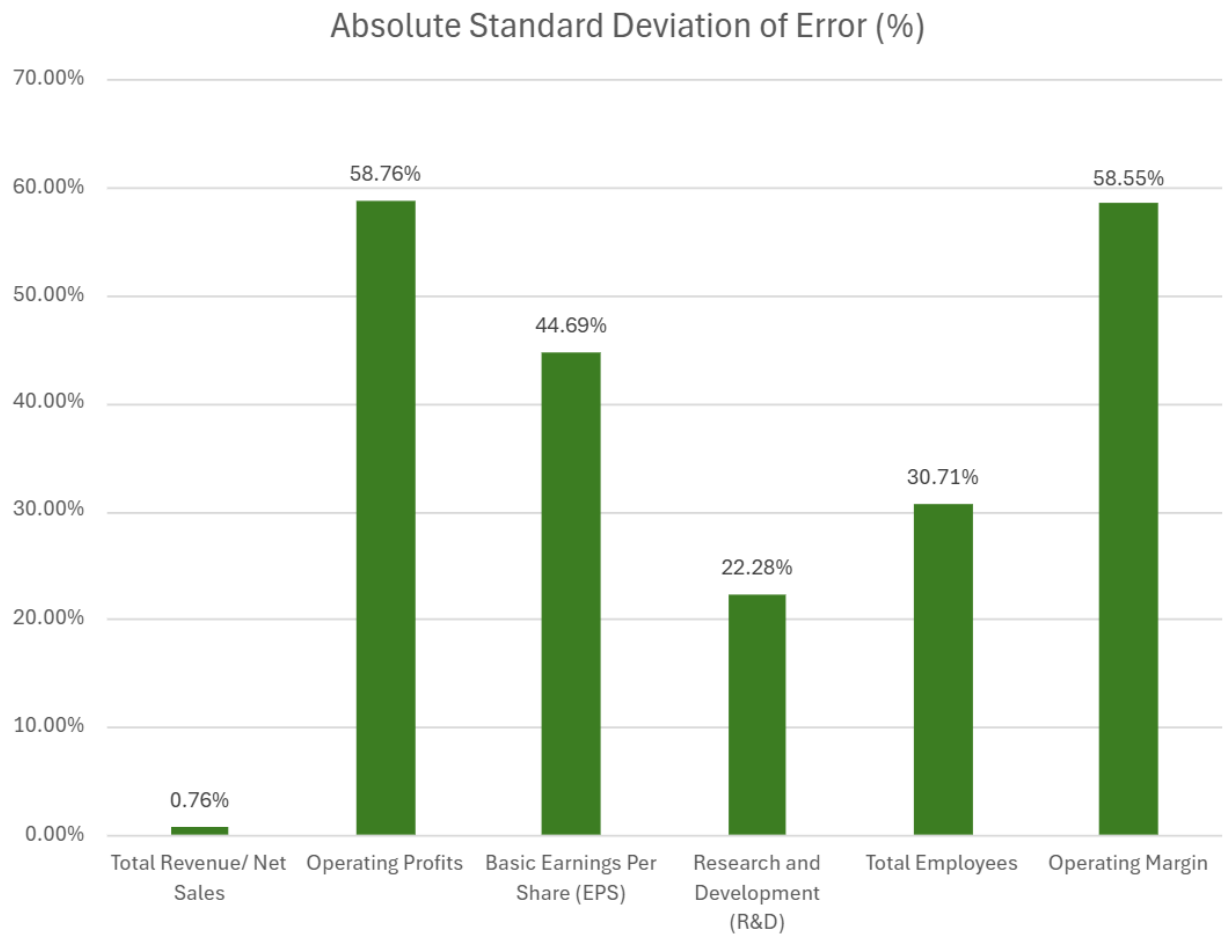


Fig. 5.

Figure 5 shows the standard deviations of errors for respective prompt questions: Total Revenue/Net Sales (0.76%), Operating Profits (58.76%), EPS (44.69%), R&D (22.28%), Total Employees (30.71%), and Operating Margin (58.55%). The average error rate for EPS was 10.156% with a higher standard deviation of 44.69%, compared to total employees with an average error rate of 10.284%. This indicates that while average error rates and standard deviations are similar, they are not proportionally consistent. This trend is also observed for operating profits and operating margin, possibly due to significantly inaccurate values in specific prompts.

For the absolute median error of each prompt question, the result was 0.00%, confirming many exact matches.

Directional k-Percentile Results

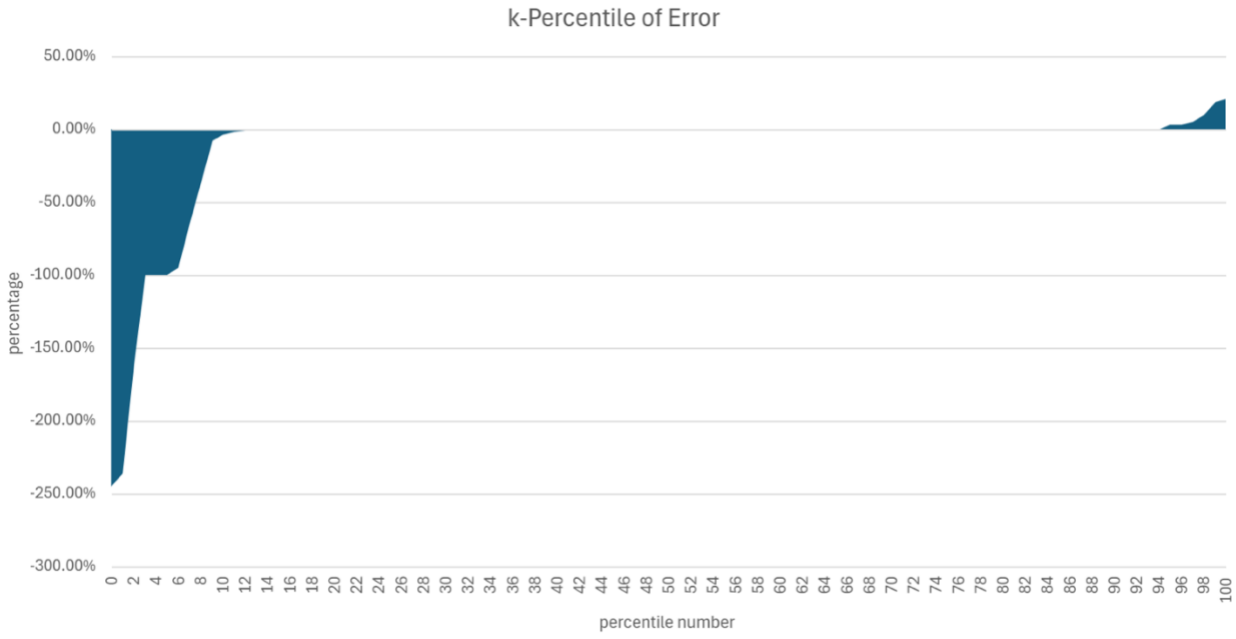


Fig. 6.

The minimum percentage error deviation observed was -245.54%, indicating a significant underestimation by GPT-4o in certain cases. The errors accounting for AI values less than their expected values occur between the 0-21st percentile (22%) while the maximum percentage error deviation observed was 20.85%. The errors accounting for AI values greater than their expected values occur between the 89-100th percentile (12%).

Underestimation error deviations less than 0% started at the 21st percentile downwards, while error deviations less than -1% occurred from the 11th percentile downwards at -1.75%. Error deviations less than -5% occurred from the 9th percentile downwards. This indicates a high accuracy rate, with 10% of the observations (0th-9th) having underestimations less than -5% (-5% is the baseline for underestimations in this study). This analysis reveals that 90% of the AI observations in terms of underestimated error are highly accurate, with errors of less than -5% in cases where the expected values are less than the actual values.

For overestimation, error deviations more than 0% start at the 89th percentile. More than 1% occurs from the 95th percentile upwards (>3.12% in deviation). Overestimations of more than 5% start at the 97th percentile(>5.56%). This indicates that only 3% of the AI observations have overestimations greater than 5% (5% is the baseline for overestimations in this study). 96% of the AI observations in terms of overestimated error are highly accurate, with errors less than 5% in cases where the expected values are more than the actual values.

Absolute k-Percentile Results

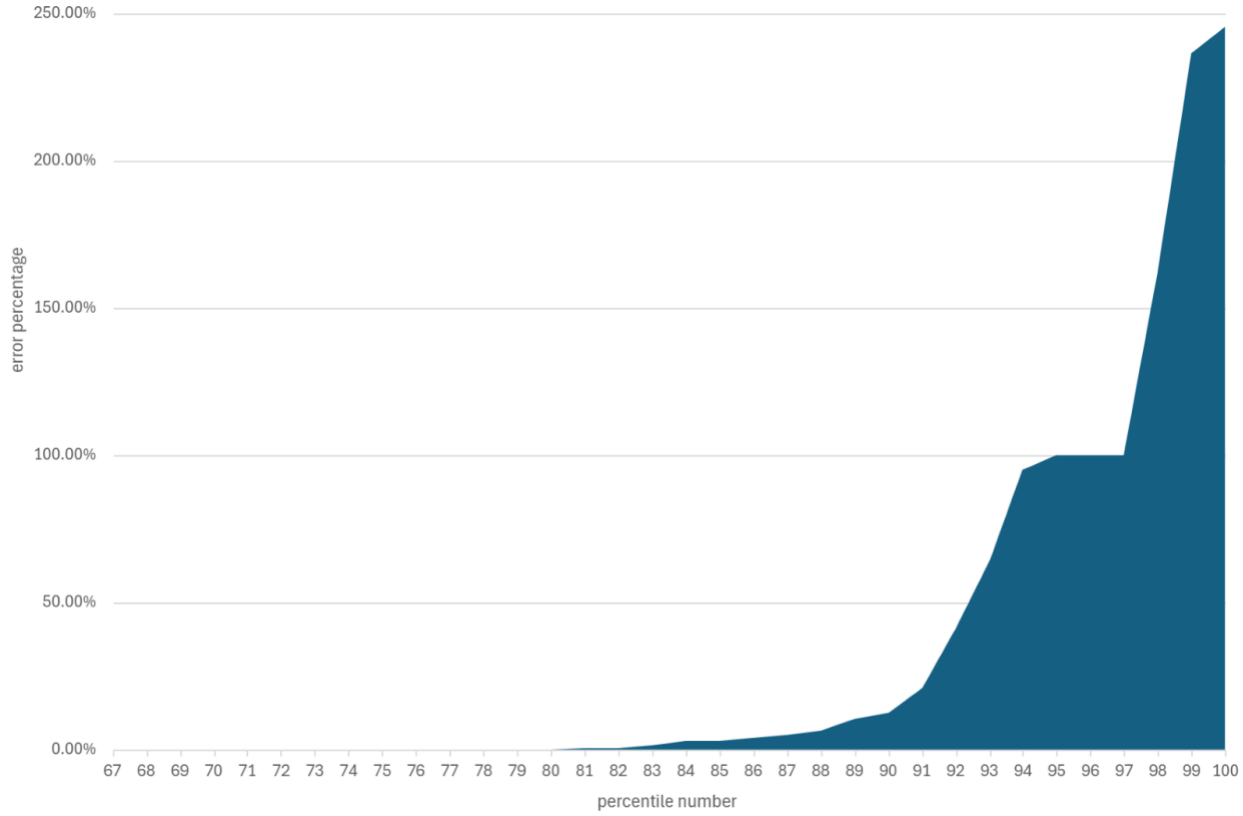


Fig. 7.

To further analyze the performance of GPT-4o in reviewing pharmaceutical annual reports, I conducted an absolute k-percentile analysis (min at 0%, max at 245.54%). This analysis helps in understanding the distribution of errors across the full sample by focusing on the magnitude of errors without considering their direction. The absolute percentage error starts at the 67th percentile, which corresponds to an error of 0.0003%, this indicates that most of the predictions have very low errors, with two-thirds of the observations having errors less than this value.

For the threshold for significant errors greater than 1% begin at the 83rd percentile. Specifically, the value at the 83rd percentile is 1.62%. This means that only 17% of the predictions have an absolute error exceeding 1%, suggesting a high level of accuracy for the majority of the data points.

The analysis shows that 86% of the predictions fall within a highly accurate range (the baseline of errors are below 5%). This high percentage of accurate predictions underscores the reliability of GPT-4o in analyzing the sample.

Summary

The comprehensive analysis of GPT-4o's performance in analyzing pharmaceutical annual reports provides several key insights into its accuracy and efficiency with a speed ratio of 5.86,

coupled with a generally high level of accuracy, underscores the potential of AI tools in accelerating data analysis processes in the pharmaceutical industry.

In our study, we analyzed 240 observations (120 for each of manual and AI datasets) and identified 40 errors. The error distribution was as follows: less than 1% (47%), between 1% and 10% exclusive (18%), between 10% and 100% exclusive (15%), and greater than or equal to 100% (20%). This indicates that a significant majority (65%) of errors are less than 10%, demonstrating a high accuracy level for GPT-4o. However, the presence of some significant errors (greater than 100%) highlights areas where the AI tool's performance can be further improved.

The analysis of errors provides a detailed look into the accuracy of GPT-4o. For directional errors, the average was -10.60%, the median was 0.00%, and the standard deviation was 41.66%. For absolute errors, the average was 11.78%, the median was 0.00%, and the standard deviation was 41.34%. These results show that GPT-4o's predictions are, on average, within 11.78% of the actual values, with half of the errors being minimal. The k-percentile analysis further revealed that 86% of the predictions are within 1% of the actual values, and 87% are within 5%, indicating high overall accuracy. Directional analysis indicated that 91% of predictions have underestimations within -5%, and 96% have overestimations within 5%, reflecting high accuracy in both directions. These findings suggest that while GPT-4o offers a compelling combination of speed and accuracy, further refinement is needed to reduce significant errors and improve overall performance.

Relevance and Implications

Firstly, the use of GPT-4o to analyze extensive pharmaceutical data has the potential to indirectly enhance drug discovery and patient care. While GPT-4o can process data faster than manual methods, allowing researchers to quickly identify trends, patterns, and anomalies, the direct impact on patient care is more subtle. This accelerated data processing can aid in the discovery of new drugs by identifying promising areas of research more efficiently. Additionally, with high accuracy rates and minimal errors, the data used for critical decisions in drug development is more reliable. Faster and more accurate data analysis can help pharmaceutical companies streamline the drug development process, potentially bringing new treatments to market sooner, which could benefit patient outcomes over time.

Secondly, this research serves as an inspiration for further studies aimed at enhancing data management practices. The integration of AI tools like GPT-4o in data analysis demonstrates the transformative potential of technology in managing large datasets efficiently. Researchers and data scientists can build on these findings to develop improved algorithms and data processing techniques that can handle even more complex datasets with greater accuracy. This can lead to the creation of more sophisticated AI models that not only analyze data but also predict future trends and outcomes, thus providing a robust framework for proactive decision-making in various sectors, including healthcare, finance, and beyond.

Lastly, the results of this study provide valuable feedback for AI developers working to refine and improve AI models. The identification of significant errors, particularly those greater than 100%, highlights areas where GPT-4o and similar models can be improved. By understanding the specific contexts and types of data that lead to these errors, developers can fine-tune their algorithms to minimize discrepancies and enhance overall performance. The detailed breakdown of directional errors offers insights into the nature of overestimations and underestimations, guiding developers in adjusting the model's parameters to achieve better balance and accuracy. Continuous feedback from real-world applications like this research ensures that AI technologies evolve to meet the growing and diverse needs of their users more effectively.

Limitations

Despite the significant findings and contributions of this research project, there are notable limitations that must be acknowledged. These limitations pertain primarily to the scope of data and the constraints of the comparative analysis among other factors.

Firstly, the scope of the data analyzed in this research is limited to annual reports from 20 pharmaceutical companies, focusing on six specific financial and operational questions. While this sample provides a snapshot of GPT-4o's performance, it is not exhaustive and may not fully represent the vast and diverse landscape of the pharmaceutical industry. The selected companies might not encompass the entire range of financial statuses, geographical locations, and market conditions that exist within the industry. Consequently, the findings might be biased towards the characteristics of these companies, limiting the generalizability of the results to a broader set of pharmaceutical firms or other sectors.

Secondly, the comparative analysis conducted in this research does not account for context-specific factors that can significantly influence data accuracy. Factors such as variations in accounting practices, differences in regulatory environments, and unique market conditions can all affect the accuracy of data interpretation and analysis. GPT-4o's performance might vary considerably when applied to data from companies operating under different regulatory frameworks or in diverse economic contexts. This limitation suggests that while GPT-4o demonstrates high accuracy in this specific study, its performance might differ when faced with varying data contexts, thereby necessitating further studies to explore these variables.

The methodology of using one prompt and response per company conversation poses limitations. This approach assumes that a single interaction is sufficient to capture the necessary data accurately, which might not always be the case. Complex financial data often requires iterative querying and cross-referencing to ensure accuracy and comprehensiveness. Relying on a single prompt could lead to oversights or misunderstandings, especially if the prompt does not fully encompass the complexity of the data being analyzed.

Another limitation is the assumption that the manual review observations are correct. The research presupposes that manual reviews are error-free and serve as a gold standard against which GPT-4o's performance is measured. However, human errors are inevitable, and manual reviews can be subject to biases, oversight, and inconsistencies. By not accounting for potential

errors in the manual reviews, the study may overestimate the accuracy of the GPT-4o when, in reality, discrepancies could be due to inaccuracies in the baseline manual data.

Future Directions

The results of this research project present several exciting avenues for future exploration and development. A key future direction is to improve the accuracy of AI tools like GPT-4o. Although the current study shows good accuracy, there is still room for improvement. This can be done by adding more advanced error correction algorithms and machine learning techniques to better understand and fix errors in data analysis. Future research should focus on finding specific areas where errors often happen and creating strategies to enhance precision or using feedback loops where human analysts check AI-generated data can help gradually improve the model's performance.

A second primary future direction involves expanding the dataset to include a more extensive range of companies and a greater diversity of financial and operational metrics. By analyzing annual reports from a larger number of pharmaceutical companies, including small and medium-sized enterprises alongside major corporations, the research can achieve a more representative understanding of GPT-4o's performance across different contexts. Additionally, incorporating more variables and questions in the analysis will allow for a more comprehensive evaluation of the AI's capabilities, capturing subtler aspects of financial and operational performance.

Another promising direction is to explore cross-disciplinary applications of the research findings. The methodologies and insights gained from this study can be applied to other fields, such as healthcare, biotechnology, and chemical engineering. By adapting the AI models to analyze data from various disciplines, researchers can assess the versatility of GPT-4o and similar tools in different scientific and technical contexts. This cross-disciplinary approach can lead to innovative applications and enhanced data-driven decision-making across multiple domains.

Expanding the scope of AI analysis to other industries beyond pharmaceuticals is a crucial step forward for future research. Industries such as finance, manufacturing, and energy also generate extensive annual reports and other financial documents that can benefit from AI-driven analysis. By applying GPT-4o and other advanced models to these sectors, researchers can evaluate the effectiveness of AI in different industrial environments and identify industry-specific challenges and opportunities. This broader application will help to validate the generalizability and utility of AI tools across diverse economic landscapes. Researchers could also consider the analysis of non-financial reports, such as environmental, social, and governance (ESG) reports, sustainability reports, and research and development (R&D) reports. These documents contain critical information that can provide insights into a company's ethical practices, innovation potential, and long-term strategic planning. By extending the analysis to these broader industry reports, researchers can develop a more holistic view of corporate performance and the impact of AI on comprehensive data assessment.

Another significant future direction involves the comparative analysis of different AI models. Evaluating the performance of GPT-4o against newer models such as GPT-5, BERT, and Google

Flan T5 can provide valuable insights into the evolution of AI capabilities. This comparative approach will help identify strengths and weaknesses of various models, guiding the selection of the most appropriate tools for specific tasks. Furthermore, it can highlight areas where current models fall short and where future improvements are needed with the use of more advanced statistical measures to provide a clearer understanding of AI accuracy and reliability, enabling more precise and actionable insights.

Conclusion

One of the most notable conclusions is the substantial time savings achieved by using GPT-4o. The AI model is approximately 5.86 times faster than manual analysis, representing a significant improvement in processing speed. This efficiency can greatly benefit the pharmaceutical industry, where timely analysis of financial data is crucial for strategic decision-making. Moreover, the AI tool demonstrated a high level of accuracy, with an accuracy rate of 86% (baseline 5%, not using direction). This indicates that GPT-4o can reliably produce results that are within a small margin of error from the actual values, making it a valuable tool for initial data analysis but despite the overall high accuracy rate, the research also highlighted a significant inaccuracy rate of 14%. This gap, particularly in the context of financial documents, is substantial and cannot be overlooked. Financial data requires a high degree of precision, and even small errors can have major implications for decision-making and regulatory compliance. The presence of some significant errors (greater than 100%) further underscores the need for caution when relying solely on AI for financial analysis. These findings suggest that while GPT-4o is highly efficient, there are still critical areas where its performance needs improvement to ensure complete reliability.

This research concludes that a combined approach, leveraging both human expertise and AI capabilities, offers the best solution for data analysis in the pharmaceutical industry. While GPT-4o can significantly reduce the time required for initial analysis, manual review is essential for verifying and refining the AI-generated results. This hybrid approach ensures that the efficiency benefits of AI are maximized without compromising the accuracy and reliability of the analysis. By integrating AI tools into the data analysis workflow, pharmaceutical companies can achieve more efficient and accurate results, ultimately enhancing their decision-making processes and strategic planning.

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